

STAT 423 Winter 2026 - Final Project

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2026-02-23

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orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(rating=orig_data$overall_rating, age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified", orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
                             ifelse(grepl("Country", orig_data$cdr_genre), "Country",
                                     ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
                                             ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
                                                     ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy, danceability = orig_data$danceability)

#Wind
cleaned_data$has_winds <- ifelse(orig_data$clarinet, "True",
                                ifelse(orig_data$flute_piccolo, "True",
                                        ifelse(orig_data$harmonica, "True",
                                              ifelse(orig_data$kazoo, "True",
                                                    ifelse(orig_data$ocarina, "True",
                                                            ifelse(orig_data$saxophone, "True",
                                                                  ifelse(orig_data$accordion, "True", "False"))))))))

#Brass
cleaned_data$has_brass <- ifelse(orig_data$trumpet, "True", "False")

#Percussion
cleaned_data$has_percussion <- ifelse(orig_data$piano_keyboard_based, "True",
                                      ifelse(orig_data$bongos, "True",
                                              ifelse(orig_data$cowbell, "True",
                                                    ifelse(orig_data$handclaps_snaps, "True", "False")))))

#Vocal or Whistle
cleaned_data$has_vocal_or_whistle <- ifelse(orig_data$vocally_based, "True",
                                             ifelse(orig_data$falsetto_vocal, "True",
                                                     ifelse(orig_data$human_whistling, "True", "False")))
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#String
cleaned_data$has_strings <- ifelse(orig_data$bass_based, "True",
                                   ifelse(orig_data$guitar_based, "True",
                                           ifelse(orig_data$orchestral_strings, "True",
                                                   ifelse(orig_data$banjo, "True",
                                                           ifelse(orig_data$mandolin, "True",
                                                                   ifelse(orig_data$pedal_lap_steel, "True",
                                                                           ifelse(orig_data$sitar, "True",
                                                                               ifelse(orig_data$ukulele, "True",
                                                                                   ifelse(orig_data$violin, "True", "False")))))))

cleaned_data$year <- as.numeric(substr(orig_data$date, 1, unlist(gregexpr("-", orig_data$date))[1] - 1))

cutoff <- round(min(cleaned_data$year) + ((max(cleaned_data$year) - min(cleaned_data$year)) / 2))

first_subpop <- cleaned_data[cleaned_data$year < cutoff, ]
second_subpop <- cleaned_data[cleaned_data$year >= cutoff, ]

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